

Output: Original String: Welcome to computer programming

Replaced String: Welcome to XYZ

3. SUBSTRING

Making our exit from the string function replace, its divisions and the usages of the various divisions, we would now stop by the string function substring. The function of the string function substring is to extract the characters from a string of two specified indices, the outcome of which is the new substring.

The characters in the programming method are addressed as “start” and “end” and are structured as follows:

```
String.substring(start, end)
```

Parameters:

Start: it stands for the characters from which the extraction is to begin. The index starts at 0.

End: not included but it stands for the place where the extraction has to stop.

```
var str = "Holla India!";  
var res = str.substring(1, 4);
```

When “start” is greater than “end”, this method will swap the two arguments, `str.substring(1, 4) == str.substring(4, 1)`. In case, “start” or “end” is less than 0, it is treated as if it were 0.

4. REVERSE

You may be demanded to reverse a string without using built-in methods or using recursion. Experts have devised ten different methods to reverse a string four of which are demonstrated below:

i) Using Algorithms

```
Function reversestring(str) {  
Return str; }  
reverseString("hello");
```

ii) Reversal with built-in functions

Though not used lately, or generally preferred by programmers, it would not be a waste to learn reversal of strings with the usage of built-in functions. There are 3 methods: `split`, `array prototype`, and `array prototype join` method.

- The `split()` method splits a string object into an array of strings by separating the string into multiple substrings.
- The `reverse()` method reverses an array in place. The first array element becomes the last and vice versa, i.e. the last array element becomes the first.